

Heterogeneous Networks on Multiple Levels

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Abstract. Heterogeneous networks and multi-level networks occur in several application fields where their integration, combination, comparison, and visualization is one of the big challenges. In this chapter, we analyze the general characteristics of this type of data and identify examples in the three application domains: biology, social sciences, and software engineering. Conceptually, we focus on sets of multivariate networks at two or more levels. Each level may describe a specific scale, and within each level several related heterogeneous networks are represented. We allow $n : m$ mappings within the same level, but only $1 : n$ mappings across levels that must be consecutive. This leads to a structured data set that is the basis for further visual analysis. Our chapter ends with ideas to visualize those networks together with the relationships between them and highlights research challenges.

Keywords: Multivariate network visualization, biological networks, social networks, networks in software engineering, multiple network analysis, multi-level networks, heterogeneous/multimodal network analysis, interaction, graph drawing

1 Introduction

At any moment in time, we are driven by and are an integral part of many interconnected, dynamically changing networks. Our species has evolved as part of diverse ecological, biological, social, and other networks over thousands of years. As part of a complex food web, we learned how to find prey and to avoid predators. We have created advanced socio-technical environments in the shape of cities, water and power systems, street, and airline systems. In 1969,

people started to interlink computers leading to the largest and most widely used networked communication infrastructure in existence today: the Internet.

Often, the complex structure of networks is influenced by system dependent local constraints on the node interconnectivity. Node characteristics may vary over time and there might be many different types of nodes. The links between nodes might be directed or undirected, and might have weights and/or additional properties that might change over time. Many natural systems never reach a steady state and non-equilibrium models need to be applied to characterize their behavior. We are in desperate need to measure, model, visualize, understand, and manage the structure and function of networks in various domains.

As a result from this usually domain-specific complexity, analysts are not only confronted with large multivariate networks. In practice, those networks can be assigned to different *levels (or scales)*, and it is absolutely possible that several different networks share the same level. In case that a set of networks share the same level, several notions of those networks can be found in the literature: they reach from *multimodal networks* to *heterogeneous networks*. For simplicity, we use the latter term in the rest of this chapter.

Such complex structures of heterogeneous networks distributed over various levels lead to a considerable visualization and analysis problem. First, the possibly tremendous size and complexity (in terms of higher dimensionality of the node/edge attributes) of those networks form a challenge of itself that is also discussed by other chapters of this book. Second and more important for this chapter, there are versatile relationships between the networks and/or between network elements even over the levels. In many application fields, it is essential to get a detailed understanding of such structures. In systems biology, for instance, networks are *the* key concept to structure and combine data that was generated by so-called high-throughput analysis methods. They can be arranged within a hierarchy of levels, from molecular-biological networks to evolutionary networks. The molecular-biological level, for example, contains a set of heterogeneous networks such as metabolic and gene regulatory networks. For a biologist it is interesting to see how the different elements in those networks are connected with each other. Here, also multivariate data plays an important role as the network elements carry additional multidimensional information, such as experimental data that changes over time.

In terms of visualization, we want to navigate and explore through this world of networks (sometimes also called *network of networks*), and a visualization tool should also provide techniques for the visual analysis of the multivariate data sets together with the underlying network topology. The upper part of Fig. 1 provides an overview of these concepts.

Many interactive visualization approaches and tools were developed for the visual analysis of graphs/networks; the same is true for the visualization of multivariate data sets. Due to the chapter format, we cannot give a comprehensive overview of the literature. Instead, we refer to textbooks, such as [24,46,47], surveys [14,23,29] and the introduction chapter of this book. Note that the given literature references only point to selected example works and make no claim

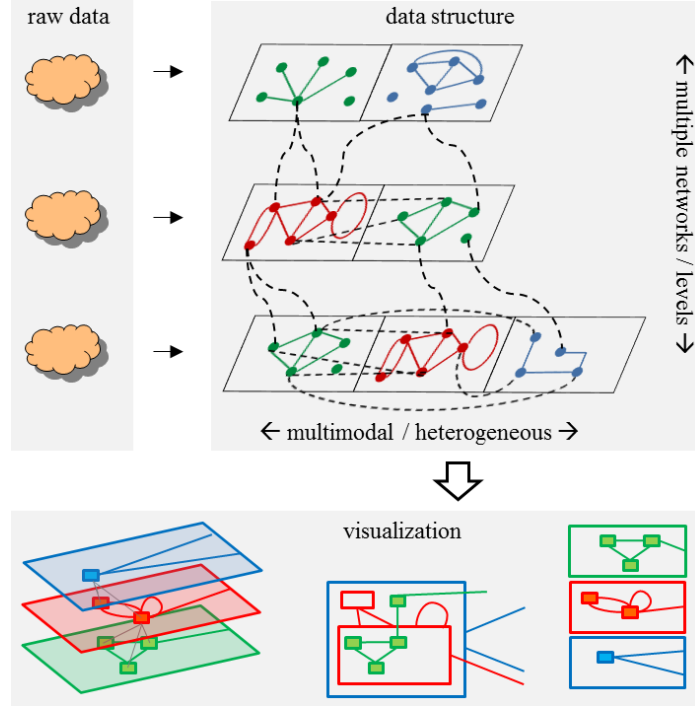


Fig. 1. A sketch of the various topics of this chapter. Colors in the top right part are used to separate different graphs (e.g., within one level), colors in the bottom part separate different levels.

to be complete. However, a number of visualizations were especially developed for the analysis of multivariate networks [19]. Some of them are based on coordinated views, such as Jigsaw [45] or enRoute [36], others provide integrated solutions, for instance MobiVis [41], GEOMI [9], or ViNCent [22,48], and there are also approaches which realize so-called attribute-driven topologies like JauntyNets [20] and GraphDice [3]. We refer the reader back to Chapter 1 for a more detailed discussion of existing directions in multivariate network visualization. To the best of our knowledge there are no efficient visualization approaches for heterogeneous networks distributed over various levels. The aim of this chapter is to formalize and highlight the underlying problems and challenges by means of three application domains as well as to propose different possibilities of solutions.

The remainder of this chapter is organized as follows: Section 2 provides a more formal specification of the data structures used in the chapter. Then, Section 3 exemplifies the visualization and analysis challenges by means of three important application fields: biology, social sciences, and software engineering. As there are almost no efficient visualization tools for multiple networks over levels available, we provide some ideas and visualization challenges in Section 4

(cp. the lower part of Fig. 1). We conclude in Section 5 and suggest some investigatory paths for future developments.

2 Formal Description of Used Data Structures

In order to facilitate the understanding and description of the following sections, we introduce a formal specification of the data structures used (a complete data structure sample is shown in Fig. 2). For this, let $G = (V, E, L)$ be a labeled graph with a finite set of nodes⁶ $V = \{v_1, \dots, v_n\}$, a finite set of edges⁷ $E = \{(v_i, v_j) | v_i, v_j \in V\}$, and a finite set of node and edge labels $L = \{l_1, \dots, l_p\}$. Each node or edge of the graph is required to have a not necessarily unique label, and $l : V, E \rightarrow L$ gives the label for each node or edge. The label is used to encode a vector of additional node-/edge-specific data (i.e., the multivariate attributes).

Let $G_1, \dots, G_m$ be a set of labeled graphs with $G_i = (V_i, E_i, L_i)$. Each graph may be directed, undirected, or mixed, and represents a specific network of the application domain (note that a graph might also be unconnected). For example, in the biological domain (see Sec. 3.1) a gene regulatory network may be represented by a directed graph, a protein interaction network by an undirected graph, and a metabolic network by a directed bipartite graph, respectively (cf. the green colored graphs in Fig. 2).

As mentioned in the introduction, we allow networks arranged at various levels. For modeling this property, let $S = \{s_1, \dots, s_k\}$ be a set of consecutive levels. Each level can contain several graphs of $G_1, \dots, G_m$, but each graph belongs to only one level. A level therefore groups graphs. The function $s : G, V, E \rightarrow S$ gives the level for each graph, node, or edge.

To connect graphs with each other, we introduce mappings between nodes of different graphs. Let $M = \{(v_i, v_j) | v_i \in V_i, v_j \in V_j\}$ be a mapping which connects a node from graph G_i with one node from a different graph G_j . Note that mappings within a graph are not allowed (those “intragraph mappings” are represented by the normal edges). Furthermore, the resulting structure could be seen as a new (global or union) graph G_G which merges all nodes and edges from $G_1, \dots, G_m$ and the mappings (edges) M .

For simplicity, the mapping M will be restricted in the following way: for $m = (v_i, v_j)$ with $v_i \in V_i$ and $v_j \in V_j$: if both nodes belong to the same level ($s(v_i) = s(v_j)$; see the red links in the figure) there will be no restriction. However, if both nodes belong to different levels (green links), a mapping is only allowed if the following two conditions hold:

1. both levels are consecutive (neighboring) levels, i.e., $s(v_i) = s(v_j) - 1$ or $s(v_j) = s(v_i) - 1$, and

⁶ Often called *vertices*; therefore the variable name v has been established to describe a node.

⁷ Also called *links*.

2. there is a $1 : n$ mapping from the higher to the lower level, i.e., if $s(v_i)$ is the higher level ($s(v_i) = s(v_j) + 1$) then there is no other node v_k in level $s(v_i)$ with $(v_k, v_j) \in M$.

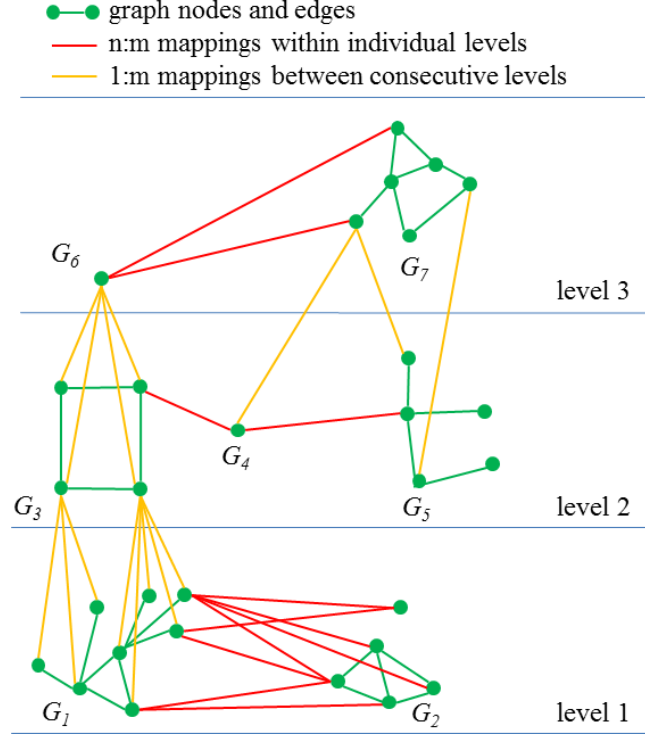


Fig. 2. The data structure described in the Sect. 2. It shows three levels with two heterogeneous networks G_1, G_2 in level 1, three heterogeneous networks G_3, G_4, G_5 in level 2, and two heterogeneous networks G_6, G_7 in level 3.

3 Application domain

3.1 Life science / Biology

Introduction Biological processes are commonly represented as networks. Examples of biological networks are molecular biological networks such as protein interaction networks (showing the interaction possibilities of proteins) and metabolic pathways (representing the transformation of metabolites into other metabolites), food webs and ecological networks (showing the dependencies between prey and predators), and phylogenetic networks (representing the evolutionary relationships between species).

A better understanding of these networks helps in making sense of much of the complex data which is nowadays available in the life sciences. Visualization is a key method to foster such understanding, and biological network visualizations therefore exist since long times. Visualization and visual analysis of these networks is an important task in life science which is evidenced by the large number of books, tools and databases which either contain manually produced drawings of biological processes and networks, or provide algorithms for their automatic layout. Many tools are available, and comparison of different tools are presented, for example, in [13,27,39]. Well known tools are

- *BiNa* [28] (<http://bit.ly/y6ix9i>)
- *BioUML* [26] (<http://bit.ly/yIETIt>)
- *CellDesigner* [12] (<http://bit.ly/A0FQiF>)
- *CellMicrocosmos* [44] (<http://bit.ly/WJ8cnE>)
- *Cytoscape* [43] (<http://bit.ly/wY2sbG>)
- *Omix* [7] (<http://bit.ly/zL52vB>)
- *Ondex* [25, Chapter 5] (<http://bit.ly/AetZjz>)
- *Pathway Projector* [27] (<http://bit.ly/zo5x2M>)
- *PathVisio* [17] (<http://bit.ly/zunwxW>)
- *SBGN-ED* [5] (<http://bit.ly/17m7KfW>)
- *Vanted* [18] (<http://bit.ly/Aigr0T>)
- *VisAnt* [15] (<http://bit.ly/agZBni>)

Data sources Biological networks may be directly derived from experimental data (such as protein interaction networks) or be build based on knowledge (such as metabolic networks). There are many data sources for biological networks: from databases covering a specific domain for a specific species (such as AraCyc [34] for metabolism in Arabidopsis) to a specific domain for a set of species (such as MetaCrop [40] for metabolism in crop plants) to several domains for several species (such as KEGG [21] for metabolic and signalling processes in a wide range of species). Another important criteria is the quality of the data which can range from completely manually curated, high-quality data to computationally derived uncurated data. For overviews of databases for a range of data domains see, e. g., [2,10].

To support exchange between tools and databases, a few standard representations are widely used such as SBML [16] and BioPAX [6]. Also the graphical representation of cellular processes and biological networks has been standardized with SBGN [30] which helps in understanding the complex processes due to the unambiguous use of glyphs. For details see the specifications of the SBGN languages [31,33,35].

Network types and examples Several types of graphs are used to represent biological networks. Typical examples are: gene regulatory and signaling networks (directed graphs) which describe how genes can be activated or repressed and therefore which proteins are produced in a cell at a particular time.

Protein interaction networks (undirected or mixed graphs) which represent the interaction between proteins such as the building of protein complexes and the activation of one protein by another protein. Metabolic networks (hyper-graphs or bipartite graphs) which show how metabolites are transformed, for example, to produce energy or synthesize substances. Phylogenetic trees (trees) which are commonly built on information from molecular biology such as DNA or protein sequences and which represent the ancestral relationships between different species. Food webs (directed graphs) which model the relationships between species in an ecological system. There is a hierarchy of biological networks, see Figure 3 and also Chapter 2.

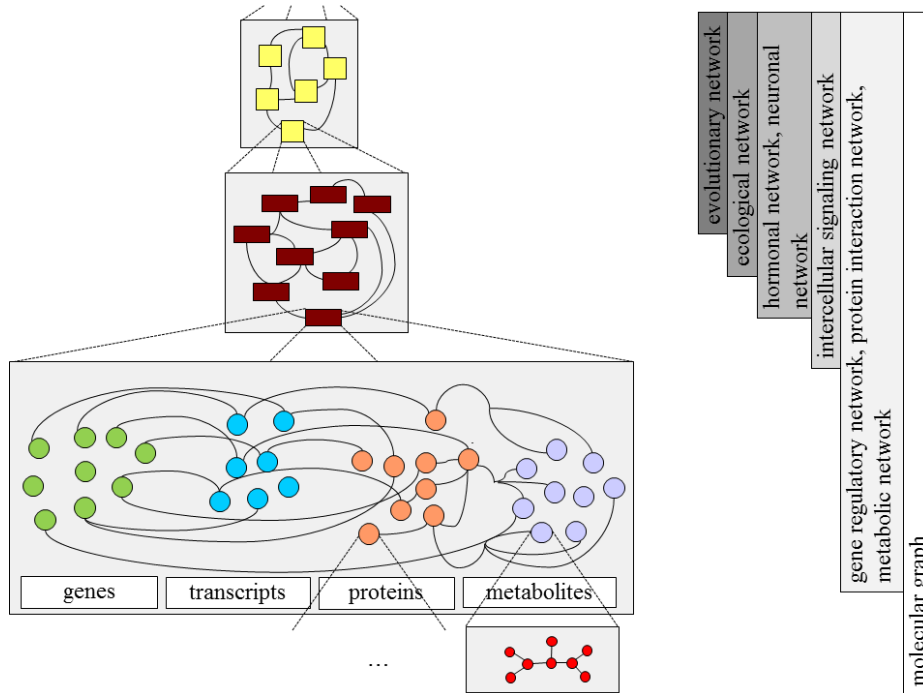


Fig. 3. A hierarchy of biological networks.

An example for an instance of our data structure described in Section 2 is given in Figure 4. Here, three molecular-biological networks (gene-regulatory network, protein interaction network, and metabolic network) are presented on level 1. Genes in the gene regulatory network may activate or inactivate the transcription of other genes. Genes are transcribed into proteins, therefore the protein interaction network does not only contain edges between nodes of the network (for representing interaction), but also (red) edges between genes (gene regulatory network) and proteins (protein interaction network). Finally the metabolic

network is a bipartite graph consisting of metabolites (circles) and enzymes (rectangles). Enzymes are proteins, therefore there are (red) edges connecting the protein interaction graph with the metabolism graph. On the next level two networks are represented: a association of all genes to their chromosomes and a clustering of all proteins into disjunct clusters.

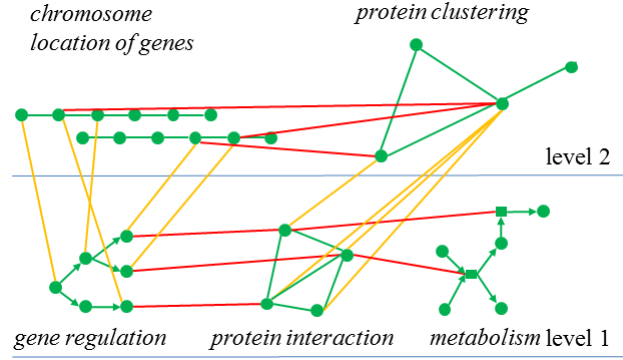


Fig. 4. An example instantiation of the data structure described in Section 2.

Use cases Here we discuss use cases that involve the multi-level heterogeneous network shown in Figure 4.

Genes encode proteins, proteins have many functions including catalyzing metabolic reactions in the form of enzymes. This major flow of information motivates the use of the tree networks on level 1 in Figure 4: gene regulatory, protein interaction, and metabolic network. These networks focus on different aspects of the same underlying biological system. In addition, level 2 gives further information which is either experimentally obtained (such as the location of genes on the chromosomes) or computed (such as the clustering of proteins based on connectivity information).

Typical tasks involving these networks are

- Which elements of the networks are important nodes (such as regulatory genes), functional building blocks (such as feed-forward motifs [32]), or relevant paths through the network (such as the shortest path between two metabolites)? Such questions are usually answered using methods from network analysis: centrality analysis to evaluate the importance of nodes or edges in the network, network clustering which may structure the network into functional modules, shortest paths between nodes representing potentially preferred routes, and so on. The result of such analysis has to be visualized in the network context and may give new insights into how important a specific gene is for the organisms or which metabolic pathway may be preferred by the organism.

- Are genes with close proximity on the chromosome are involved in the same processes? For example, if genes in close proximity on the chromosome are regulated by the same regulators in the gene regulatory network or if they fall in the same cluster in the protein clustering then they are likely to belong to the same biological process.
- Is experimental data in agreement with knowledge represented by the networks or not? Lots of experimental data can be attached to nodes and edges: experimental conditions such as treatments or temperature; time series measurements; and so on. For example, the following attributes can be used: edge weights (e.g., to represent fluxes), node weights (e.g., to show concentrations), node presence/absence (e.g., to model knockouts), and node shape (e.g., to show different types of biological objects). Multivariate data has first to be integrated into the biological network. Data mapping deals with this integration of additional data into networks. An examples is the integration of metabolomics, transcriptomics, and fluxomics measurement data into the metabolic network. This data can be mapped on different network elements (such as metabolites, enzymes, and reaction edges). The mapped data should behave in a way which could be explained by the underlying networks. If, for example, in the gene regulatory network only some of the depended genes get activated if a regulatory gene is active, this could be an indication for an yet unknown additional regulatory process.

Challenges Tools for the visualization of analysis results often provide standard interaction techniques such as

1. Zooming (sometimes also semantic zooming)
2. Filtering
3. Collapsing and expanding of structures
4. Highlighting

There are many tool available, but choosing the right tool is often difficult (for some more widely used tools see the introduction of this Section). Standard visualization tasks are often sufficiently solved, but more elaborated visualization, interaction, and layout are an open issue. An overview concerning open problems in biological network visualization can be found in [1].

3.2 Networks in Social Sciences (4-5 pages, Katy)

This section reviews networks in the social sciences, with a special focus on multi-level scholarly networks, drawing on research in the sociology, information science, statistics, graph theory, and network science. We first present a domain overview introducing relevant terminology along the way. The next subsection discusses major data sources and formats. The third subsection discusses diverse network types and interlinkages together with examples. Subsection four discusses concrete use cases that benefit from the formalization presented in section two and the visualization solutions discussed in section four. We conclude with a discussion of challenges for multi-level network analysis and visualization in the social sciences.

3.2.1 Introduction

Social networks exist at the individual (micro) to population (macro) levels. Examples are networks of friendship and hate, collaboration and competition, trade and blockage between entity nodes such as individuals, organizations, cities, countries, geospatial regions, or areas of science. Typically, an entity node is part of multiple types of networks, e.g., a person is part of friendship, collaboration, and family relationship networks. Nodes might contribute to information diffusion networks by receiving and sending emails, tweets, or commenting on digital objects. Nodes from different levels interact and influence each other, e.g., individual authors of papers might be aggregated at the department or even country level to analyze expertise profiles and collaboration patterns of departments or countries. The movement of a highly cited author from one departments or country to another will impact both: expertise profiles (entity properties) and collaboration patterns (entity linkages) of associated entity nodes. In other cases, networks from different levels are nested, e.g., a person is part of local social networks that can be further aggregation to global, population level networks. Last but not least, the structure of multi-level networks impacts the utility of the network for, e.g., information diffusion. The stronger the edge between two nodes is, the more information can flow; the more often two nodes share information, the stronger their edge grows over time.

Understanding the evolution of network structure and dynamics has far ranging applications. Among them are the prevention of disease spreading, e.g., identifying highly connected individuals that should be vaccinated first when trying to fight a pandemic or using social networks to reducing the social diffusion of smoking or bad eating behaviors (see subsection 3.2.4); to increase the spreading of information, e.g., in education or when designing effective (viral) marketing strategies; to help manage the extremely complex decision making space of professional career choices (see subsection 3.2.4); or to form and sustain productive research and development teams.

3.2.2 Data Sources and Formats

Social network data might be qualitative or quantitative. Qualitative data is commonly acquired via surveys, interviews, direct observation, or by reviewing written documents. Quantitative data might be derived from existing databases (e.g., phone address data revealing social networks; publication data for extracting co-author networks) or acquired via measurement. Common data sources are social media data (phone, email, blogs, Twitter, Facebook), scholarly data (scientific publications, patents, funding

Kommentar [KB1]:

cite
[Börner, Katy, Soma Sanyal](#), and Alessandro Vespignani. 2007. "Network Science". In *Annual Review of Information Science & Technology*, edited by [Cronin, Blaise](#), Ch. 12, 537-607. Medford, NJ: Information Today, Inc./American Society for Information Science and Technology.

awards), or custom data collected in research studies. Commonly studied networks are social networks, friendship networks, collaboration networks, political networks, trading networks, citation and other knowledge diffusion networks.

Nodes and their linkages can be represented as an adjacency matrix, an edge list, or lists of nodes and edges. Consequently, most network science tools¹ support Pajek .net or edge list formats as well as Graph/ML, XGMML formats. Many tools support the extraction of networks from common file formats, e.g., publication data formats from the Web of Science, Scopus or personal bibliography tools such as EndNote or LaTeX's bibtex, or from generic tabular formats, e.g., .csv files.

3.2.3 Network Types and Examples

This subsection discusses different network types using examples from scholarly network analysis—the study of authors and their papers as we assume all readers are familiar with this data.

Networks might be directed or undirected, weighted or unweighted, valued or not. Linkages might be among units of the same type, such as friendship or coauthorship linkages, or between units of different types, such as authors and the papers they produce. In general, three types of linkages are distinguished: *direct linkages* such as paper citation linkages; *co-occurrence* linkages of words or references; and *cocitation linkages*, e.g., of authors or papers. Plus, units of the same type can be interlinked via different link types; for instance, teenagers might be linked via love and hate relationships; papers can be linked according to cword, cocitation, or bibliographic coupling analysis. Linkages might be directed and/or weighted. Each nonsymmetrical occurrence matrix has two associated (symmetrical) co-occurrence matrices; for instance, for each paper citation matrix exists a bibliographic coupling and a cocitation matrix.

An example of direct linkages are paper-paper (citation) linkages: Papers cite other papers via references, forming a nonweighted, directed paper citation graph. It is beneficial to indicate the direction of information flow from older to younger papers via arrows. References enable readers to search the citation graph backward in time. Citations to a paper support the forward traversal of the graph. Citing and being cited can be seen as two vital roles of a paper.

Co-occurrence linkages interconnect co-author networks. Having the names of two authors (or their institutions and countries) jointly listed on one paper, patent, or grant is an empirical manifestation of scholarly collaboration. The more often two authors collaborate, the greater the weight of their joint coauthor link. Weighted, undirected coauthorship networks appear to have a high correlation with social networks that are themselves shaped by geographic proximity.

Word co-occurrence linkages are used to calculate the topic similarity of basic and aggregate units of science. Units that share more words are assumed to have higher topic overlap and are connected via linkages and/or placed in closer proximity on a topic map. Co-occurrence networks are weighted and undirected.

¹ <http://sci2.wiki.cns.iu.edu/display/SCI2TUTORIAL/8.2+Network+Analysis+and+Other+Tools>

Cocitation linkages: Two basic or aggregate units of science are said to be cocited if papers associated with them jointly appear in the list of references of a third paper. The more often two units are cocited, the more they are expected to have something in common. Examples are document cocitation and author co-citation networks.

Given a data file with publication records retrieved from the Web of Science or Scopus database, more than 30 different networks can be extracted. Some of these networks, e.g., co-author or paper-citation networks have been studied extensively and their structure, distribution, and evolution are known. Other network types—particularly heterogeneous networks interlinking different levels—have not yet been studied in detail.

3.2.4 Use Cases

This subsection exemplarily discusses two use cases that involve multi-level heterogeneous networks. For each, we identify user needs/tasks, workflows, and insights gained.

Reducing Social Contagion of Smoking Behavior

Social peer pressure is powerful. The desire of an individual to be an accepted member of a group frequently leads to behavior that the individual would not show without the group. For example, it is well known that—all other things being equal—joining a well-organized team of experts will lead to higher professional performance than joining a team of less skilled, less organized individuals. Studies by Christakis and Fowler have evaluated a densely interconnected social network of 12,067 people assessed repeatedly from 1971 to 2003 as part of the Framingham Heart Study to show the relevance of social networks in the diffusion of smoking, obesity, but also loneliness. Relevant news stories were entitled: “Are Your Friends Making You Fat?” While it is difficult and in some cases impossible to distinguish correlation vs. causation, e.g., some of the shown effects could be due to the “birds of a feather flock together” factor also known as homophily, networks seem to impact outcomes. A scientific approach that considers homophily, environment, and induction, e.g., by using multi-level heterogeneous networks, seems promising.

For example, it seems desirable to study individuals in the context of multiple, interconnected networks as shown in Figure 3.1. Level 1 exemplarily depicts different social networks of a teenager represented by node A. Among them are family, school, and out-of-school friendship networks. Level 2 shows aggregations of these networks to families and clans (groups of teenagers) that have diverse profiles and interlinkages. While there is no Level 3, geography appears to matter: The ease by which A can come into contact with teenagers and other individuals that have of different demographic profiles is important. Using this representation, the number of linkages to positive and negative influences can be calculated. Possibly, the rate of contact and the geospatial and peer pressure on an individual can be determined (but see 3.2.5 Challenges). Using computational models, it might be possible to predict the general impact of network changes on individual behavior.

Fact is, people are connected to each other and to their environment; changes in social networks or in the environment will likely impact their behavior. The big, open question is: How to change social networks and/or environments to cause positive change?

Kommentar [KB2]: The Spread of Obesity in a Large Social Network over 32 Years

Nicholas A. Christakis, M.D., Ph.D., M.P.H., and James H. Fowler, Ph.D. N Engl J Med 2007; 357:370-379 July 26, 2007 DOI: 10.1056/NEJMs066082

Kommentar [KB3]: <http://www.nytimes.com/2009/09/13/magazine/13contagion-t.html?pagewanted=all&r=0>

Kommentar [KB4]: cite Detecting implausible social network effects in acne, height, and headaches: longitudinal analysis

BMJ 2008; 337 doi: <http://dx.doi.org/10.1136/bmj.a2533> (Published 4 December 2008) Cite this as: BMJ 2008;337:a2533

and

Lyons paper on "The Spread of Evidence-Poor Medicine via Flawed Social-Network Analysis," published in the journal *Statistics, Politics, and Policy*,

Kommentar [KB5]: We could place A in Level 2 and use Level 1 to represent geography (networks). Does any of the other domains have a need for 3d space?

Designing Successful Career Trajectories

Pathways that individuals take from taking their very first job and then moving to the next during their working years are called career trajectories. They might be plotted over time, geospatial space, or topic space. Individuals might change locations and jobs because of warfare, political problems, or ethnic purging or because of voluntary or volitional migration where individuals choose to relocate to new places because of opportunities offered in the new place. In the latter case, social factors (e.g., closeness to family and friends, standard of living), colored by cultural, historical, linguistic, or weather considerations, but also active encouragement by visa and immigration controls are key criteria. Education is another major factor—low education typically equates low paying jobs and little resources to pay for education or to move to a different place. High education commonly leads to better paying jobs and the generation of financial wealth that pays for personal education or the education of family members. Highly specialized expertise profiles might also mean that only few jobs exist that match this expertise profile and international migration is required to find an appropriate job.

Coming back to the study of scholarly networks, much data exists to track career trajectories of scholars over time. The U.S. National Science Foundation has been conducting the Survey of Doctorate Recipients (SDR) biennially since 1973. The SDR follows a sample of U.S. trained doctorates in science, engineering, and health fields throughout their careers from shortly after degree award by a U.S. institution through age 75. Detailed information on professional position, salary, number of kids, etc. is available for each respondent in a longitudinal fashion. In addition, there exist extensive publication and funding databases that record funding intake (number of awards and funding amounts) and publication output (number of papers and their citations) over time together with information on co-investigators and co-authors but also institutional affiliations, and acknowledged grant funding that interlinks funding and publications.

In general, career trajectories are best viewed as decisions over time. As it is much easier to change institutions than to change a topical area of expertise; researchers change geolocations frequently (particularly in the begin of their career) but are less likely to change their topic area and if so decide to venture into topically similar areas of science that benefit from the same skill set/expertise.

Figure 3.2 shows the three level heterogeneous network of a scholar node B and associated family links, mostly local friendship networks, (international) co-author relations, collegial networks at the same institution. Family and friendship links, the reputation of another institution, but also attributes of the new geolocation (living costs, weather, etc.) have the power to make a person look for or accept another job—if a job offer is made. The new institutional environment together with new friendship and collaboration opportunities will change funding and publication patterns and ultimately the conditions for the next career step.

Not only individuals and families but also companies and organizations migrate in response to market changes, to maximize economic utility, or to be close to customers.

Kommentar [KB6]: Again, I am inclined to use Level 1 to represent geospatial features—possibly as a network, e.g., air traffic as a proxy for reachability.

Level 2 for social + collab networks

Level 3 for networks/reputation of institutions and scientific disciplines.

3.2.5 Challenges

The extraction of heterogeneous networks requires the identification of unique entities, e.g., people, institutions, scientific areas, and their interlinkage via (different types of) edges. While publications, patents, and other digital documents have unique digital object identifiers; the development of digital object identifiers for authors, institutions, and scientific areas is under active development but not yet usable. Similarly, the interlinkage of (heterogeneous) networks across levels poses serious unification and linkage challenges. Many problems require $n:m$ mappings not only within one level but also between consecutive levels—implications for restricting the mapping to $1:m$ are unknown.

Few algorithms exist to study heterogeneous networks—most algorithms assume there is one type of nodes and one type of edges. Visualizations are important to communicate complex heterogeneous, multi-level networks. However, most readers never learned how to read a network and seeing multiple interlinked networks might be overwhelming.

3.2 Software Engineering

Introduction A plethora of tasks in software engineering involves analyzing software artifacts that are best represented using graphs. The graphs can be time-dependent ($n:m$ mapping within a level), e. g., data from cvs, svn, or other repositories, and multi-level ($1:n$ mapping between levels), e. g., package structure in Java. Further, they have with different node and edge types, and to each of these node and edges additional information can be associated. Thus, the graphs have all the properties described in Section 2 and shown in Fig. 2.

Data sources The data produced and used in software engineering is manifold. First of all, software development processes can be mapped to directed graphs with loops, where each development step is mapped onto a node, and an edge designates the sequence of these steps. Further, software development artifacts like system designs, detailed designs, and the static and dynamic structure of programs can be mapped onto directed and undirected graphs. Finally, algorithms can be visualized. In this case, graphs are more rarely used, except that the algorithm requires the handling of graphs.

Network types and examples Software processes are mapped onto graphs similarly to other processes, e. g., business processes, schedules, or production processes. Each development step is mapped onto a node and can have additional information assigned, e. g., number of developers, time allocated for this step, a list of tasks, inputs, and deliverables. Each step is connected to the following step by an edge. This edge normally is guarded in the sense that it can be only followed if the deliverables of the previous step are ready (to a certain amount). As the processes can branch and can have loops, the graphs can not be reduced to trees or DAGs (directed acyclic graphs). Mostly, the process comes in an abstract form that is instantiated for a specific project. Therefore, several similar graphs exist that can be compared to each other, e. g., the instantiations among each other. Further, the process can be in the form of a hierarchy. Then, each node of the high level process is refined into a number of steps that can be refined themselves.

The static structure of software artifacts can also be mapped to typically directed graphs. Taking a typical java source code, these graphs have the following properties. The hierarchy of java packages can be mapped to different levels. The lowest level graph nodes can be either classes or methods, members, and subclasses of classes. The snapshots in time (typically saved in revision control systems) form a sequence of hierarchical graphs. Many revision control systems allow branches and merges, such that the sequence becomes a directed acyclic graph. Possible links (edges) on class level are inheritance, method call, aggregation, implements relation, and usage relation (similar to aggregation, where the class is used as type of another class's member, it can also be used as type of return values, parameters, or exceptions). Besides these directed links (edges), also undirected links (edges) might exist like code clones, common fate (evolution, classes that are often changed together), or semantic similarity. Aggregating

the classes and sub-packages in packages directly implies the aggregation of the edges; thus, a hierarchy is built for one time-step.

The dynamic structure of a software system also leads to a sequence of graphs. However, these graphs typically have a 1:n mapping as the underlying static structure does not change. Instead, individual edges are activated and deactivated according to the dynamic evolution of the graphs. Indeed, the nodes of the dynamic graphs are instances of the nodes of the static graph, where each static node can lead to an arbitrary number of dynamic nodes. The same holds for the edges: call graph edges get instantiated during the evolution of the run. Of course, the granularity can be again on class or on any of the package levels.

Use cases Use cases are typically directly derived from the software engineering tasks. Task that benefit from visualizations include

- Software understanding: Understanding the functionality and the interplay of components necessitates the understanding of the static and dynamic structure. Further, it is the basis for the maintenance task.
- Maintenance: To maintain a system, a thorough understanding of the individual components as well as their dependencies is needed. The dependencies are best analyzed based on visualizations of the static structure and dynamic runs.
- Re-engineering: Re-engineering is needed for legacy systems or whenever the new developer can not directly access the knowledge of the original developers. Like Software understanding, the interplay of the different components plays an important role, which is best understood using static and dynamic graphs.
- Testing and bug fixing: For bug fixing, the interplay of components is very important, because side effects on other components should be minimized when changing an individual component. The same holds for adding new functionality. The current behavior should not change by adding additional functionality.
- Product Lines: Product lines incorporate basic functionality in a core of components, adding functionality by additional or changed components. For product lines, it is mandatory to understand the relationship between the different products. The goal is, to maximize the amount of common components, minimizing the effort for creating additional functionality using additional components.

Challenges There exist a lot of software visualizations but only few of them are scalable and comprehensive. Most of the existing solutions either focus on specific properties of the software artifact, like metrics, static structure (UML diagrams) at one time step with several types of edges or static structure over time with one edge type.

But what is really necessary for program comprehension is to provide integrated views, like the AreaView tool by Telea et al. While this tool integrates

UML diagrams with metrics and several areas of interests, it does not display neither hierarchies nor several time steps. What is needed is an integrated set of visualizations and interactions that allows to look at static software artifacts from several points of view showing or filtering information on demand to allow a focused analysis of the software artifacts. Such a set of interactive visualizations has to take into account graphs with the following properties:

1. Time-dependent static structures, e.g., svn or cvs snapshots
2. Structure hierarchies, e.g., packages and classes in Java
3. Distinct node types, e.g., classes and interfaces in Java
4. Distinct edge types, e.g., inheritance, aggregation, method calls (directed) and code clones, semantic similarity (undirected)
5. Software metrics as additional node information, e.g., lines of code, number of methods, depth of inheritance
6. Metrics associated with edges, e.g., number of method call relations, similarity

While all of these properties directly map to visualization properties and challenges, interaction has always to be considered as part of the solution. Standard interaction techniques

1. Filtering
2. Highlighting
3. Collapsing structures
4. Expanding structures

are mandatory. For comparing two graphs, highlighting selected nodes and edges is beneficial, e.g., if small multiples (coordinated views are used). Collapsing (parts of) a graph and expanding it allows for navigating the hierarchy (levels). Marking a node in a higher level together with the related nodes/edges in the lower levels allows for tracing structures over levels.

Only if the user is allowed to explore selected parts of the software artifacts freely, she will be able to gain new insights and find the information needed for solving specific tasks. Thus, interactive visualizations can be embedded in the software analysts workflow.

4 Visualization

The visualization of heterogeneous networks on multiple levels is still relatively unexplored in the literature. However, there are a number of visualization approaches that focus on solving specific analysis tasks or operate on a subset of the data structure introduced in Section 2, for instance on heterogeneous multivariate networks only. In the following, we provide a short overview on those techniques that might be considered as related work.

The most obvious visualization metaphor for networks on different levels is stacking. All networks on the same level are layouted (by using any more or less smart graph drawing algorithm) on a 2D plane, and then these planes are stacked

in 3D (cf. Fig. 5). Multivariate data attached to the nodes or edges might be displayed within the planes themselves, as additional layers below or above the individual 2D planes or separated into multiple coordinated views [38]. One of the earlier examples is VisLink [4] which is a general approach to show relationships between visualizations. In our special case, networks on levels are displayed on multiple 2D planes that can be arranged in the third dimension in various ways (in parallel, book-like, ...); relationships are represented as links, i.e., as inter-plane edges. Multivariate data can be represented inside of the planes or on additional individual planes. In the latter case, inter-plane edges might be used to point to the attached multivariate data. Note that in most cases—according to our definitions—heterogeneous networks are placed to different planes and not networks on multiple levels. The most obvious drawback of such 3D techniques is their low level of scalability.

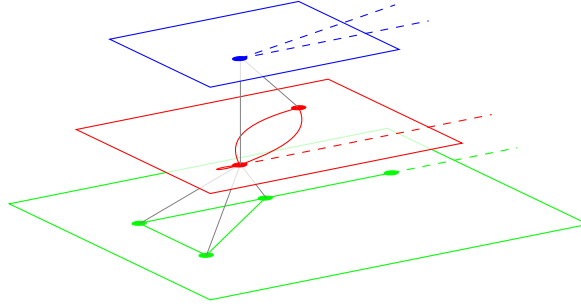


Fig. 5. Stacked 2D planes of network drawings.

Some approaches abstain from explicit encoding of the network topology and visualize aggregated information only. This idea supports the analysis of very large data sets in terms of many heterogeneous networks and large multivariate attributes. ManyNets [11] represents networks as rows in a table together with their multivariate data (primary as well as secondary data [23]) similar to the well-known TableLens [37]. Several interaction possibilities support the visual analysis of the networks which might also be displayed as node-link diagram on demand. GraphTrail [8] has similar aims, but in contrast to show the networks in a table, they have chosen to represent the network elements in an aggregated form. For doing this, standard charts like bar charts or tag clouds are employed that can be interactively arranged on a canvas. GraphTrail also supports the analysis process by providing a history functionality. Other approaches abstract directly in the node-link representations, such as OntoVis [42]: an ontology graph, which describes the node categories/clusters and their relationships, serves as vehicle to control the abstraction and navigation processes. All these tools have in common that they provide solutions to analyze a set of heterogeneous multivariate networks, but *not* at multiple levels.

TODO:

- Nesting
- Alignment

Note: The nesting sketch requires that there are no inter-edges that connect a node at Level i with two separate networks at Level $i+1$. Thus, the graph that is built by ALL inter-edges and the corresponding end-nodes over all levels and networks within the levels must form a *forest*. In the general case, it is a DAG.

TODO: challenges

5 Discussion and conclusion

(all)

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